Gabriele Baldi

Software Engineer

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Summary

Software engineer with six years of experience in industrial software, IoT and test automation. I built an Azure IoT Hub platform connecting 150+ machines and a Python test automation framework covering close to 90% of controller test cases, working closely with customers, product owners and developers, and I validated a product idea with market research. I am looking for software engineering roles with high technical impact and strong exposure to product and business.

Skills

- **Programming:** Python, C#, AI-assisted development (Kiro, Claude, Cursor)
- **Cloud and IoT:** Azure Cloud, MQTT, OPC UA, FastAPI, Linux
- **Testing:** Robot Framework, TestRail, Test strategy
- **Application development:** WPF/XAML, MVVM, MVC, Desktop and web applications, API integration

Experience

QA Engineer (Test Automation), Omron Industrial Automation Europe

Technical consultant with Randstad Digital, placed at Omron · Nov 2025 to present · Barcelona

- Built from scratch a Python and Robot Framework test automation framework, adopted by the team for regression and controller testing, covering close to 90% of controller test cases and integrated with the TestRail API.
- Defined an AI-assisted test automation workflow, from test design to TestRail reporting, and documented it as a proposed team standard.
- Partnered with the product owner, project team lead and developers from the user stories onward, defining what to test around user value.

Software Engineer, Paolino Bacci

Nov 2023 to Aug 2025 · Cascina, Italy

- Architected and deployed an Azure IoT Hub platform connecting 150+ industrial machines for real-time monitoring and remote control (MQTT, OPC UA).
- Worked directly with customers to commission machines running my software, turning their feedback into product updates.
- Built multi-threaded desktop applications for machine control (WPF/XAML, MVVM) and 3D simulation and visualization tools.
- Researched cloud solutions and competitors in woodworking manufacturing to support the company's digitalization strategy.

Software and Robotics Engineer, ESA Automation

Jul 2022 to May 2023 · Vicopisano, Italy

- Developed application software for industrial robots (kinematics, vision, HMI, communication protocols) and programmed ABB and KUKA robots.

Software Engineer, Scuola Superiore Sant'Anna

Feb 2020 to May 2022

- Developed control software for a diagnostic automation proof-of-concept machine (C++, Arduino, PLC Structured Text), collaborating with the electronics team on PCB design and debugging.
- Built a real-time diagnostics dashboard (ThingsBoard, FastAPI) on an on-premises Linux server.

Projects

- **Energy monitoring web platform (freelance, 1 year).** For a small Italian industrial company, built a web application with an admin configuration page for remote monitoring of energy and consumption devices. Worked directly with the client, shaped the UX around his feedback and tuned the platform on one of his customers. Github Project
- **MyMeal, mobile app to reduce household food waste.** Validated the market before building: 130+ survey responses and customer conversations, personas and jobs-to-be-done, TAM/SAM/SOM and competitor analysis, pitch deck and business plan, then

built an Android prototype. MyMeal

Education

- **MSc Mechatronic Engineering**, Politecnico di Torino (2017 to 2019)
- **BSc Automation Engineering**, Università degli Studi di Napoli Federico II (2011 to 2017); Erasmus exchange at Universidad de Sevilla (2014 to 2015)

Languages

Italian (native) · English (professional working proficiency, Cambridge FCE B2) · Spanish (full professional proficiency)

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